

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Discrete Mathematics & Set Theory	
Course code	CS1812	
Semester:	I	
Names of the Course Instructor(s)	Dr. Rekha Satish Dr. Shrikant Udagi Dr. Manjunath S	
Course credit	No. of hours per week (2+1+0)	Total no. of Teaching hours
03	03 Hours	45 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/ Reference
1	1	Sets and operations on sets, Set-builder notations, types of sets, union, intersection	Lecture + PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
2	1	complement, difference, symmetric difference, their properties and laws, Venn diagrams	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
3	1	Tutorial.	Discussion + Examples	
4	1	Cartesian Product, cardinality; multisets	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
5	1	Tutorial	Discussion + Examples	
6	2	Propositions, Application and Importance of Logic and Logical Operators	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications

7	2	Propositional logic(continue)	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
8	2	Tutorial	Discussion + Examples	
9	2	Tautology, contradictions	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
10	2	Logical Equivalence	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
11	2	Tutorial	Discussion + Examples	
12	2	Predicate Logic	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
13	2	Predicate Logic	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
14	2	Tutorial	Discussion + Examples	
15	3	Relations	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications

16	3	Functions—one-to-one, onto functions	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
17	3	Tutorial	Discussion + Examples/ Activity	
18	3	The Pigeon-hole Principle	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
19	3	Function Composition	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
20	3	(Contd.) The Pigeon-Hole Principle, Function composition	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
21	3	Tutorial	Discussion + Examples	
22	3	Function Composition, invertibility and inverse of functions	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
23	3	(Contd.) Function Composition, invertibility and inverse of functions	Discussion + Examples/ Activity	Rosen K, Discrete Mathematics and Its Applications
24	4	Introduction on Permutations	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications

25	4	The Rules of Sum and Product	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
26	4	Tutorial	Discussion + Examples	
27	4	Problems on Permutations with repetitions	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
28	4	Permutation(Continue)	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
29	4	Tutorial	Discussion + Examples	
30	5	Combinations	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
31	5	Combinations(Continue)	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
32	5	Tutorials	Discussion + Examples	
33	5	Combinations with repetition	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications

34	5	Combinations with repetition(Continue)	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
35	5	Tutorial	Discussion + Examples	
36	5	Binomial Theorem	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
37	5	Binomial theorem and Problems	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
38	5	Tutorial	Discussion + Examples	
39	5	Application of Combination and Binomial Theorem	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
40	5	Catalan Number	Lecture + chalk and Talk/ PPT/Activity	Rosen K, Discrete Mathematics and Its Applications
41	5	Tutorial	Discussion + Examples	Rosen K, Discrete Mathematics and Its Applications
42		Activity	Octave/ Practice	

43		Activity	Octave/ Practice	
44		Activity	Octave/ Practice	
45		Activity	Octave/ Practice	

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component (Marks)
- CIE 1-A (Tutorial)	10
- CIE 1-B (Poster Presentation)	10
- CIE 2	25
- CIE 3-A (Tutorial)	10
- CIE 3-B (Octave Activity)	15

Semester End Exams (SEE)	
Mode of Exam	Pen and Paper
Weightage of exam	30

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
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CO 1	Apply the concepts of set theory to solve problems in relevant domains.	3	2			2			2		2		
CO 2	Apply truth tables, logical equivalences, and predicate logic to solve reasoning-based problems	3	2			2			2		2		
CO 3	Apply relations, functions, and the pigeonhole principle to solve computational problems	3	2		2	2			2		2		
CO 4	Apply principles of combinatorics to solve basic counting problems.	3	2		2	2			2		2		

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Tutorial 1 Graded Component 1	Tutorial will be conducted for 10 marks.
	Poster Presentation	Poster Presentation for 10 Marks.

	Graded Component-2	
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Graded Component-1 (10 Marks):

Poster Presentation rubrics (Max: 10 M)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1	Creativity, On time submission (4 Marks)	Observations	Highly creative design; original approach; submitted before/on deadline. (4 Marks)	Some creativity; submitted on time but lacks originality. (2–3 Marks)	Very basic or copied design; submitted late. (0–1 Marks)
2	Content (4 Marks)	Observations	Accurate, comprehensive, and relevant content; well-structured. (4 Marks)	Mostly relevant with few minor inaccuracies or missing elements. (2–3 Marks)	Incomplete, unclear, or inaccurate content. (0–1 Marks)
3	Viva and Documentation (2 Marks)	Viva Voce Observations	Clear explanation during viva; well-organized documentation. (2 Marks)	Some explanation with basic documentation. (1 Mark)	Unclear explanation or missing documentation. (0 Marks)

Graded Component-2 (10 Marks):

Tutorial Rubrics (Max: 10 M)					
1	Correctness of Answers (10 Marks))	Answer Sheet Evaluation	All answers are correct; logical, complete work shown. (10 Marks)	Most answers are correct; minor calculation or conceptual errors. (6–8 Marks)	Several errors or incomplete/missing answers. (0–4 Marks)

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	The test will be conducted for 25 marks(Pen and

		Paper) and will be evaluated according to the Scheme of Evaluation prepared by team of course faculty.
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Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Tutorial 2 Graded Component 1	Tutorial will be conducted for 10 marks.
	Octave Activity Component 2	Octave Activity for 15 Marks.

Graded Component-1 (10 Marks):

Tutorial Rubrics (Max: 10 M)

1	Correctness of Answers (10 Marks))	Answer Sheet Evaluation	All answers are correct; logical, complete work shown. (10 Marks)	Most answers are correct; minor calculation or conceptual errors. (6–8 Marks)	Several errors or incomplete/missing answers. (0–4 Marks)
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Graded Component-2 (15 Marks):

Octave Activity Rubrics (Max: 15 M)

1	Correctness of answers (4 Marks	Observations	All answers are correct and logically justified. (4 Marks)	Most answers are correct; minor errors present. (2–3 Marks)	Several incorrect answers; lacks understanding. (0–1 Mark)
2	Ontime Submission (1 marks)	Observations	Submitted on or before the deadline. (1 Mark)	Submitted slightly late (within grace period). (0.5 Mark)	Submitted after the deadline. (0 Mark)

3	viva (10 marks)	Viva Voce	Excellent understanding and confident explanation; uses Octave effectively. (9–10 Marks)	Moderate understanding; explains partially with some guidance. (5–8 Marks)	Poor or no understanding; unable to answer or use Octave. (0–4 Marks)
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Signature of the Course Faculty